Hugo Marquerie 04/01/2026

Teo con ceros polinomios espacio afin ideal propio cuerpo alg cerrado no vacio

Teorema 1. *Sea K un cuerpo algebraicamente cerrado y $J \subsetneq K[x_1, \dots, x_n]$ un ideal propio, entonces $\mathbb{V}(J) \neq \varnothing$.*

***Demostración*:** Como J es un ideal propio, $\exists m \subset K[x_1, \dots, x_n]$ ideal maximal tal que $J \subset m$. Por el teorema anterior, $\exists \{a_i\}_{i=1}^n : m = \langle x_1 - a_1, \dots, x_n - a_n \rangle$. Entonces, como $J \subset m$, tenemos que $\mathbb{V}(J) \supset \mathbb{V}(m) = \{(a_1, \dots, a_n)\} \neq \varnothing$, luego $\mathbb{V}(J) \neq \varnothing$. ■

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